From Compute to Productivity

Nvidia, GPU Constraints and the Economics of AI Diffusion

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Abstract

AI adoption has become widespread, while productivity effects remain uneven. This paper studies the cost between using AI tools and turning them into measurable output. The research question is how constrained access to advanced GPU compute affects the cost of productive AI adoption. The paper uses the 2026 AI Index Report and Nvidia's 2026 Annual Report to build a focused economic mechanism. Advanced GPUs are treated as an upstream input in the AI production stack. Nvidia is used as a case because its infrastructure position makes the compute bottleneck visible. The paper argues that constrained GPU access raises productive AI adoption cost through compute cost and access risk. It also explains how export controls create substitution incentives, especially in U.S.-China technology competition. The conclusion is simple. AI diffusion depends on the system that converts capability into work.

Research Roadmap

This paper follows a chain of questions. First, why can AI adoption move faster than productivity? Second, what costs appear between model access and productive use? Third, why does advanced GPU compute become a bottleneck in that conversion process? Fourth, how does U.S.-China technology competition add policy risk to GPU access?

The paper answers these questions through one mechanism. Constrained access to advanced GPU compute raises the cost of converting AI adoption into productive use. Nvidia reveals the mechanism because downstream demand for AI capability creates upstream demand for advanced compute systems. Export controls deepen the mechanism by turning access stability into an economic variable.

Introduction

Artificial intelligence is often discussed through its most visible layer, models. Public attention follows stronger reasoning systems and more capable agents. Economic analysis starts one layer deeper. A firm can adopt AI tools and still struggle to turn them into measurable productivity. A country can access models and still lack the infrastructure needed to use them at scale. The main issue is the conversion from AI capability to productive capacity.

The 2026 AI Index describes the current moment as a gap between "what AI can do and how prepared we are to manage it" (Stanford HAI, 2026, p. 3). That captures the problem this paper studies. AI capability is advancing quickly, and organizational adoption is spreading fast, yet productivity gains remain uneven across tasks and firms. The question is how costly it is to make AI useful.

This paper studies that cost through a focused economic, constrained access to advanced GPU compute. Advanced GPUs support model training and inference. They also support large-scale deployment. Nvidia is used as the central case because it reveals the bottleneck clearly. Its annual report states that AI runs on "real hardware... and real economics" (NVIDIA, 2026, p. 2). That moves AI from a software story into an input-cost story.

The research question is clear. How does constrained access to advanced GPU compute affect the cost of converting AI adoption into productive use? The argument is that AI diffusion depends on a conversion system. Compute is the first layer. Infrastructure and firm integration shape the next layers. Policy access affects whether the system can remain stable.

Literature Review

This section separates three ideas that are often treated as one. AI adoption shows that firms are trying the technology. Productive use shows whether the technology changes output. Compute infrastructure explains the cost between the two.

The first problem is the gap between adoption and productive use. The AI Index reports that generative AI reached nearly 53 percent population-level adoption within three years. Organizational adoption reached 88 percent (Stanford HAI, 2026, p. 4). These number shows that AI has moved into and ordinary work, and productivity conversion remains the hard stage.

The same report gives a reason for caution. “The data does not point in a single direction” (Stanford HAI, 2026, p. 4). This matters because adoption data can overstate economic value. A company may buy access to AI tools and remain stuck at limited deployment. The question becomes less about who has AI and more about who can make AI work.

Corporate deployment data makes the gap visible. The AI Index states that AI agent deployment remains in single digits across nearly all business functions (Stanford HAI, 2026, p. 11). This supports the starting. AI tools are spreading faster than firms which were reorganized work around them.

The second is the cost structure behind AI use. Nvidia describes modern data centers as AI factories, which convert energy into tokens. Nvidia also writes that “Cost per token defines efficiency” (NVIDIA, 2026, p. 5). This gives AI an industrial cost structure. Tokens are produced, priced and used.

Inference strengthens this cost structure. Nvidia states that inference defines the economics of AI for agentic systems (NVIDIA, 2026, p. 6). Agents require ongoing inference because they operate through multiple steps. As agentic use grows, productive AI adoption becomes tied to the cost of running systems continuous.

Existing reports describe rapid adoption and expanding infrastructure. While the cost of conversion. Has been missed. AI capability has given accessed by computer, while infrastructure and firm integration determine whether that capability becomes productivity.

Research Question and Variables

From AI adoption to productivity, is an uneven production. This paper studies one narrow mechanism inside it. How constrained access to advanced GPU compute affects the cost of converting AI adoption into productive use.

The independent variable is constrained access to advanced GPU compute, which includes physical scarcity and policy risk. 1. Physical scarcity refers to limited availability of advanced compute capacity. 2. Policy risk refers to uncertainty when chip access depends on export rules.

The dependent variable is the cost of converting AI adoption into productive use. This cost includes compute cost and integration cost. Compute cost covers the price of running AI systems. Integration cost covers the effort required to place AI inside real workflows.

The expected relationship is straightforward. Constrained access to advanced GPU compute raises the cost of productive AI adoption. This relationship treats compute as the first cost layer in a longer conversion process.

Methodology

This paper uses secondary source analysis and microeconomic modeling. The goal is to explain a mechanism, how constrained access to advanced GPU compute changes the cost of productive AI adoption.

The AI Index Report 2026 is used as the main external source, which provides evidence on AI adoption and compute capacity. Which also provides evidence on AI sovereignty. This source is useful because it separates fast adoption from uneven productivity.

Nvidia's 2026 Annual Report is used as a corporate source and case study. It explains how Nvidia defines AI infrastructure through AI factories and cost per token. This source is useful because it shows how a chip company describes the economics of AI production from inside the industry.

The paper treats Nvidia's report as a strategic company document. It reflects Nvidia's position, so it is used carefully. The AI Index Report balances it with external data.

The microeconomic model is that,

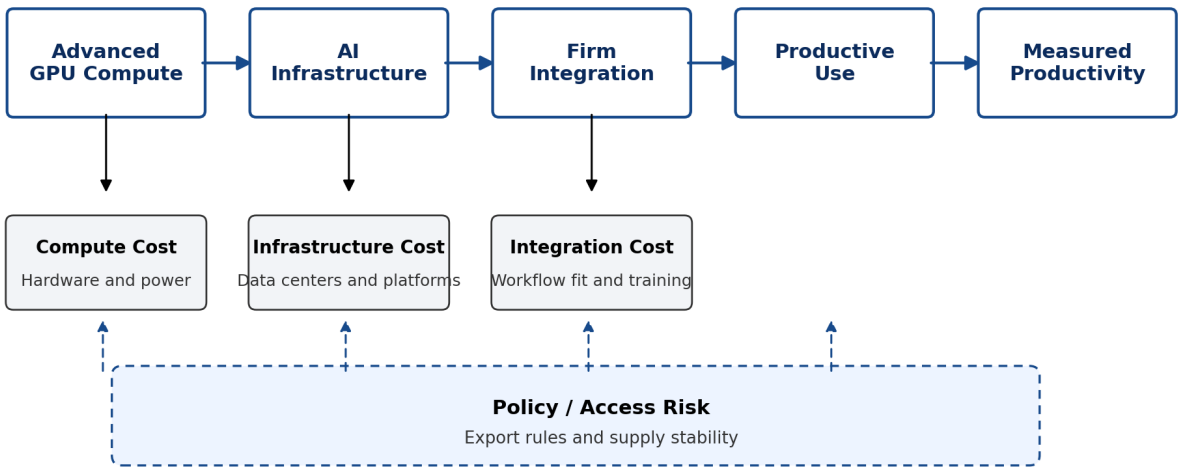
Productive AI Adoption Cost = Compute Cost + Infrastructure Cost +Integration Cost +Policy and Access Risk

GPU constraints increase compute cost; Higher compute cost raises the price of AI services; Higher service cost makes productive AI adoption harder for firms with weaker infrastructure.

The paper also uses derived demand. Firms demand GPUs because they demand AI capability; AI applications create demand for models; Models create demand for training and inference; Training and inference create demand for advanced compute.

Figure 2. Compute-to-Productivity Conversion Model

Capability becomes productivity through a conversion system.



Key insight, compute opens access; integration turns capability into output.

Analysis I, Compute as the First Layer

The first analytical explains why compute is central to AI productivity. AI systems require model access, and model access requires compute. As AI moves from simple chat to agentic workflows, the system generates more tokens and runs for longer periods; which makes compute a continuing cost.

Nvidia's annual report describes modern data centers as AI factories. These factories convert energy into tokens (NVIDIA, 2026, p. 5). The wording is useful because it gives AI a production structure. Energy as an input, and tokens are the intermediate output. Productive work appears when those tokens are used inside real tasks.

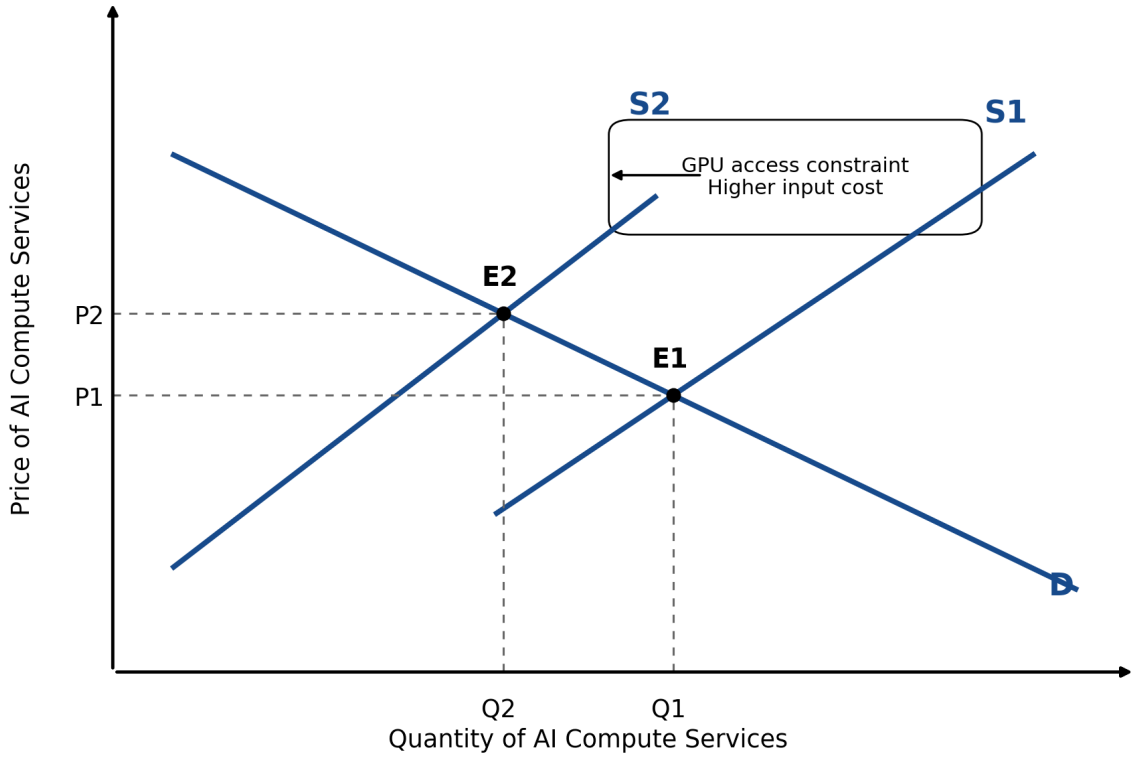
A lower cost per token makes it easier to use AI at scale. A higher cost per token makes every query and workflow more expensive. This is why compute cost belongs inside the dependent variable.

Training creates the model, while inference runs the model in daily use. Agents reason through steps, and as a result, agentic AI depends heavily on inference cause. Then, productive AI adoption depends on the cost of continuous operation.

When advanced GPU access becomes constrained, the cost of compute rises. Then, the supply of AI compute services becomes more expensive. The difference for companies survive is that, firms with stronger capital can continue scaling but smaller firms face higher barriers.

Figure 1 shows GPU constraints operate an input-cost shock. The supply curve for AI compute services shifts left. Price rises, and accessible quantity falls. Which explains the first economic barrier in the conversion process.

Figure 1. Market for AI Compute Services Under GPU Constraint



Result, price rises; accessible quantity falls.

Analysis II, Nvidia as an Input Supplier

This explains why Nvidia is the clearest case for this paper. Nvidia is treated as an upstream input supplier in the AI production stack. The economic concept is derived demand.

Firms demand Nvidia GPUs because they demand AI capability. AI applications create demand for models. Models create demand for training and inference. Training and inference create demand for advanced compute. Advanced compute creates demand for GPU systems.

Nvidia's annual report supports this. The company describes its Data Center platform as focused on compute-intensive workloads. Which also says that the platform is deployed in cloud and enterprise data centers (NVIDIA, 2026, p. 99). Nvidia sells into the infrastructure layer of AI production.

The AI Index adds an external view of concentration, which reports that global AI compute capacity reached 17.1 million H100-equivalents by the end of 2025. It also states that Nvidia accounts for over 60 percent of total compute (Stanford HAI, 2026, p. 15). Huawei holds a smaller and growing share.

A concentrated upstream input can shape the cost of downstream adoption. If a firm can access advanced Nvidia compute, it can experiment with larger AI workloads. If access is expensive or restricted, the firm faces a higher conversion cost.

Nvidia therefore matters because it reveals the structure behind AI diffusion. AI adoption at the application layer depends on compute capacity at the infrastructure layer. The more AI moves into production, the more the upstream input constraint matters.

Analysis III, U.S.-China Competition and Substitution Incentives

The third is policy risk. Advanced GPU access is shaped by market price and government restriction. This changes the cost of productive AI adoption for firms that depend on foreign compute supply.

Export controls affect the supply side first. When advanced GPUs face cross-border restrictions, the accessible supply of AI compute falls for restricted customers. In a standard supply and demand diagram, this works like an input constraint. The supply curve for AI compute services shifts left. Price rises, and accessible quantity falls.

The deeper effect appears on the demand side. A technically superior input can become a risky input when policy access is unstable. A Chinese firm may prefer Nvidia hardware for performance. However, that preference changes when supply depends on export approval.

This is where Huawei enters the economic story. The question is why customers begin to accept substitutes. Export controls make Nvidia access less predictable. Less predictable access raises the expected cost of dependence. That expected cost creates demand for domestic alternatives.

Nvidia's own risk discussion points that, the company states that export controls may disrupt supply and distribution channels. It also warns that such controls may benefit competitors with alternatives that face fewer restrictions (NVIDIA, 2026, p. 114).

The restriction therefore has two effects. It limits short-run access to advanced U.S. chips. It also gives China stronger incentives to build alternatives. A policy designed to preserve technological advantage can increase the long-run demand for domestic substitution.

This mechanism explains why Nvidia cares about Chinese alternatives. The larger risk is ecosystem fragmentation. If Chinese firms build around domestic chips, the switching cost away from Nvidia may fall over time. Nvidia's position as the default upstream supplier becomes less secure.

Analysis IV, Strategic Demand and the AI Race

The fourth explains why demand for AI compute can behave differently from ordinary demand. Firms and countries invest in AI because current productivity matters. Future position matters as well.

AI compute demand includes strategic pressure. A firm may invest before the exact return is clear because falling behind competitors has a cost. The same logic applies to countries. A country may pursue domestic compute capacity because it wants control over future capability.

This makes AI demand partly strategic demand. The buyer purchases present output and future option value. Compute gives firms the ability to experiment and scale when useful applications appear. Lacking compute can mean lacking the ability to participate later.

The AI Index describes AI sovereignty as a defining feature of national policy. Which also states that state-backed investment in AI supercomputing is rising (Stanford HAI, 2026, p. 12). This supports that states treat AI capacity as more than a private technology investment.

Nvidia's annual report reflects the same shift. It states that sovereign AI business more than tripled to over 30 billion dollars (NVIDIA, 2026, p. 14). This shows that demand for compute comes from firms and states. The shared concern is control over future capability.

The economic result is a feedback loop. More expected AI competition increases demand for compute. Higher compute demand strengthens Nvidia's bottleneck position. Policy restrictions increase the desire for substitutes. And the market is shaped by present cost and future position at the same time.

Evaluation

The argument has limits. Advanced GPU access is a major constraint, while the conversion from AI adoption to productivity also depends on model efficiency and firm learning.

Open models are the first limit. Nvidia states that open models lower barriers to entry and allow more actors to build specialized systems (NVIDIA, 2026, p. 7). This means model access can become easier even when frontier compute remains concentrated.

Open models also create a second effect. When more developers could build AI systems, token generation grows. More token generation creates more demand for infrastructure. Open models may lower the model-access barrier while increasing pressure on the compute layer.

Model efficiency is another limit. The AI Index notes that some smaller models have achieved strong results through better post-training techniques (Stanford HAI, 2026, p. 15). If smaller systems become more useful, some firms may reduce their need for the most advanced GPUs.

Firm integration remains central. Compute can support AI use, and the firm still has to redesign work. A company needs data access and clear tasks. Without these, AI remains a tool at the edge of production.

These limits refine the paper's claim. Productive AI use emerges when compute and integration work together.

Conclusion

AI diffusion often measured by adoption. This paper measures the harder stage—conversion. A firm may have access to AI tools and still fail to produce measurable value. The reason that AI capability has to pass through compute, infrastructure and firm integration before it becomes work.

Advanced GPU compute is the first layer of that conversion system. Nvidia matters because it sits close to this layer. Its GPUs and platforms support training and inference workloads. As downstream demand for AI applications rises, upstream demand for advanced compute rises with it. This is the derived demand behind Nvidia's position.

U.S.-China technology competition adds another layer of cost. Export controls make GPU access less stable for restricted customers. That instability changes incentives. Firms and countries have stronger reasons to search for domestic substitutes when reliance on Nvidia becomes a policy risk.

The main conclusion is that, constrained access to advanced GPU compute raises the cost of productive AI adoption. The effect appears through higher compute costs and greater access risk, which is strongest for firms and countries that lack their own compute infrastructure.

AI is spreading through a production system. Compute gives access to capability. Infrastructure and integration decide whether that capability becomes output—Who can make AI work.

Appendix A, Evidence Matrix

The table shows how each source supports a specific part of the argument. The goal is citation discipline. Each piece of evidence is assigned to a question and a section.

Evidence	Function	Section	Citation
AI capability gap	Shows why capability and management systems must be separated.	Introduction	Stanford HAI, 2026, p. 3
53 percent population adoption and 88 percent organizational adoption	Shows that AI adoption is already large.	Literature Review	Stanford HAI, 2026, p. 4
Data does not point in a single direction	Keeps the paper from treating adoption as automatic productivity.	Literature Review	Stanford HAI, 2026, p. 4
Agent deployment remains in single digits	Shows that productive deployment is still incomplete.	Literature Review	Stanford HAI, 2026, p. 11
AI factories convert energy into tokens	Defines compute as a production process.	Analysis I	NVIDIA, 2026, p. 5
Cost per token defines efficiency	Supports compute cost as a measurable unit cost.	Analysis I	NVIDIA, 2026, p. 5
Nvidia accounts for over 60 percent of total compute	Supports Nvidia as a bottleneck case.	Analysis II	Stanford HAI, 2026, p. 15
Data Center revenue reached 194 billion dollars	Shows downstream AI demand pulling upstream compute demand.	Analysis II	NVIDIA, 2026, p. 14
Export controls may benefit competitors with fewer restrictions	Explains substitution incentives.	Analysis III	NVIDIA, 2026, p. 114
AI sovereignty is becoming a defining feature of national policy	Explains strategic demand.	Analysis IV	Stanford HAI, 2026, p. 12
Sovereign AI business more than tripled to over 30 billion dollars	Shows that states are part of compute demand.	Analysis IV	NVIDIA, 2026, p. 14
Open models lower barriers to entry	Provides the main limitation.	Evaluation	NVIDIA, 2026, p. 7

References

NVIDIA Corporation. (2026). 2026 annual report. NVIDIA Corporation.

Stanford Institute for Human-Centered Artificial Intelligence. (2026). Artificial Intelligence Index Report 2026. Stanford University.